

# Vectors and Vector Spaces

Part 3/3: Independence and Orthogonality  
CSC 1105 Mathematics for Computer Science

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October 5, 2026

# Lecture Outline

- Linear dependence and independence
- Matrix tests for independence
- Bases and dimension
- Dot products, norms and distance
- Angles between vectors
- Orthogonal vectors

# Learning Outcomes

By the end of this lecture, you should be able to:

- Determine whether a set of vectors is linearly independent or dependent.
- Use matrix pivots, rank, and determinants to test linear independence.
- Identify a basis and determine the dimension of a span.
- Compute dot products, norms, distances, and angles between vectors.
- Determine whether two vectors are orthogonal.

## Review of Part 2

For vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ , put

$$A = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_k] .$$

- $A\mathbf{c} = \mathbf{w}$  asks whether  $\mathbf{w}$  belongs to their span.
- A basis is a linearly independent spanning set.
- Basis coordinates are unique; the number of basis vectors is the dimension.

### Question

Can different choices of coefficients produce the same vector? Independence determines the answer.

# Linear Independence

## Definition

A list  $\mathbf{v}_1, \dots, \mathbf{v}_k$  in a real vector space is **linearly independent** if

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0} \implies a_1 = \dots = a_k = 0.$$

The all-zero coefficients give the **trivial relation**.

The list is **linearly dependent** if a relation exists in which *at least one* coefficient is nonzero.

## A Dependence Relation: Example

Recall the vectors from Part 2:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Since  $\mathbf{w} = \mathbf{u} + \mathbf{v}$ ,

$$\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}.$$

The coefficients  $(1, 1, -1)$  are not all zero, so the list is dependent.

What the relation shows

The third vector adds no new direction to the span.

## An Independent Pair: Example

Let  $\mathbf{u} = (1, 2)^T$  and  $\mathbf{v} = (2, 1)^T$ . To test independence, solve

$$a\mathbf{u} + b\mathbf{v} = \mathbf{0} \iff \begin{cases} a + 2b = 0, \\ 2a + b = 0. \end{cases}$$

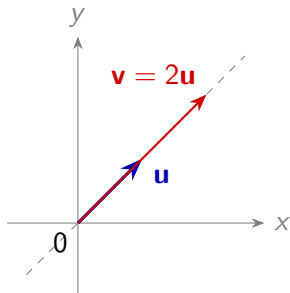
Subtracting twice the first equation from the second gives

$$-3b = 0 \implies b = 0, \quad a = 0.$$

Only the trivial relation exists. Therefore the pair is independent.

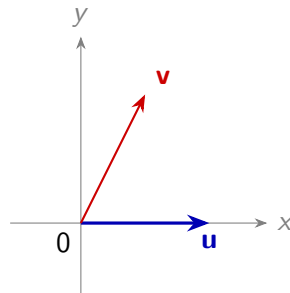
## Geometric Interpretation in $\mathbb{R}^2$

**Dependent pair**



Two nonzero scalar multiples span one line.

**Independent pair**



Two vectors that are not scalar multiples span the plane.

# Dependence and Redundancy

Suppose

$$a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k = \mathbf{0}, \quad a_j \neq 0.$$

Rearranging expresses one vector using the others:

$$\mathbf{v}_j = - \sum_{i \neq j} \frac{a_i}{a_j} \mathbf{v}_i.$$

Conversely, such an expression can be rearranged into a nontrivial dependence relation.

## Equivalent description

A finite list is dependent exactly when one of its vectors is a linear combination of the others.

## Pairwise Independence Is Not Sufficient

Consider three vectors in  $\mathbb{R}^2$ :

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

No pair consists of scalar multiples, so every pair is independent.

However, the entire list is dependent because

$$\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}.$$

Independence must be tested for the **whole list**.

# The Matrix Test for Independence

Let  $A = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k] \in \mathbb{R}^{n \times k}$ .

$$A\mathbf{c} = \mathbf{0} \iff c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}.$$

## Independence criterion

The columns are independent exactly when  $A\mathbf{c} = \mathbf{0}$  has only the zero solution: no free variables, a pivot in every column, and  $\text{rank}(A) = k$ .

| Question                             | Required pivots         |
|--------------------------------------|-------------------------|
| Are the columns independent?         | A pivot in every column |
| Do the columns span $\mathbb{R}^n$ ? | A pivot in every row    |

## A Pivot in Every Column: Example

Test  $(1, 0, 1)^T, (0, 1, 1)^T, (1, 1, 0)^T$ :

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}.$$

There are three pivots, one in every column.

In the homogeneous system, back-substitution gives

$$-2c_3 = 0, \quad c_2 + c_3 = 0, \quad c_1 + c_3 = 0,$$

so  $c_1 = c_2 = c_3 = 0$ . The columns are independent and form a basis of  $\mathbb{R}^3$ .

## Useful Independence Facts

- A list containing only one vector is independent exactly when that vector is nonzero.
- Any list containing the zero vector is dependent: use coefficient 1 for it and 0 for the others.
- More than  $n$  vectors in  $\mathbb{R}^n$  are dependent: an  $n$ -row matrix cannot have more than  $n$  pivots.
- For  $n$  vectors in  $\mathbb{R}^n$ , the columns are independent exactly when the square matrix has nonzero determinant.

The determinant test applies only to **square** matrices.

## Linear Independence: Exercise

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

- 1 Test whether  $\mathbf{u}, \mathbf{v}, \mathbf{w}$  are independent.
- 2 If they are dependent, find a nontrivial relation.
- 3 Remove a redundant vector and give a basis for their span.

# The Dot Product

## Definition in $\mathbb{R}^n$

For  $\mathbf{u} = (u_1, \dots, u_n)^T$  and  $\mathbf{v} = (v_1, \dots, v_n)^T$ ,

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

The result is a **scalar**, not a vector.

For  $\mathbf{u} = (1, 2, -1)^T$  and  $\mathbf{v} = (3, 0, 4)^T$ ,

$$\mathbf{u} \cdot \mathbf{v} = 1(3) + 2(0) + (-1)(4) = -1.$$

# Dot Product Properties

For real vectors of the same length and scalars  $\alpha, \beta$ ,

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u},$$

$$\mathbf{u} \cdot (\alpha \mathbf{v} + \beta \mathbf{w}) = \alpha(\mathbf{u} \cdot \mathbf{v}) + \beta(\mathbf{u} \cdot \mathbf{w}),$$

$$\mathbf{u} \cdot \mathbf{u} = u_1^2 + \cdots + u_n^2 \geq 0.$$

Moreover,

$$\mathbf{u} \cdot \mathbf{u} = 0 \iff \mathbf{u} = \mathbf{0}.$$

Symmetry also gives linearity in the first argument.

## Norm and Distance

The Euclidean norm and distance can be written using dot products:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}, \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

For  $\mathbf{u} = (1, 2)^T$  and  $\mathbf{v} = (4, 6)^T$ ,

$$\|\mathbf{u}\| = \sqrt{1^2 + 2^2} = \sqrt{5},$$

$$d(\mathbf{u}, \mathbf{v}) = \|(3, 4)^T\| = \sqrt{3^2 + 4^2} = 5.$$

# The Angle Between Vectors

For **nonzero** vectors, the angle  $\theta \in [0, \pi]$  satisfies

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta, \quad \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

For  $\mathbf{u} = (1, 0)^T$  and  $\mathbf{v} = (1, 1)^T$ ,

$$\cos \theta = \frac{1}{1\sqrt{2}} = \frac{1}{\sqrt{2}} \implies \theta = 45^\circ.$$

The angle with the zero vector is undefined.

# The Sign of the Dot Product

For nonzero vectors:

| Dot product                       | Angle               | Interpretation |
|-----------------------------------|---------------------|----------------|
| $\mathbf{u} \cdot \mathbf{v} > 0$ | $\theta < 90^\circ$ | Acute angle    |
| $\mathbf{u} \cdot \mathbf{v} = 0$ | $\theta = 90^\circ$ | Perpendicular  |
| $\mathbf{u} \cdot \mathbf{v} < 0$ | $\theta > 90^\circ$ | Obtuse angle   |

The magnitude also depends on the vector lengths. Dividing by both lengths isolates the angle.

# Orthogonal Vectors

## Definition

Vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$  are **orthogonal** when

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

For  $\mathbf{u} = (1, 2, -1)^T$  and  $\mathbf{v} = (1, 1, 3)^T$ ,

$$\mathbf{u} \cdot \mathbf{v} = 1 + 2 - 3 = 0.$$

They are orthogonal, but their lengths are  $\sqrt{6}$  and  $\sqrt{11}$ .

The zero vector is orthogonal to every vector. It has no direction and cannot belong to an independent list.

**Note:** Orthogonal vectors are vectors that are perpendicular to each other.

## Revision Questions

- 1 Test  $(1, 2, 0)^T$ ,  $(0, 1, 1)^T$ ,  $(2, 5, 1)^T$  for independence. If dependent, give a relation and a basis for their span.
- 2 Does  $\{(1, 1, 0)^T, (0, 1, 1)^T, (1, 0, 1)^T\}$  form a basis of  $\mathbb{R}^3$ ? Justify your answer.
- 3 For  $\mathbf{u} = (1, 2, -1)^T$  and  $\mathbf{v} = (2, 1, 4)^T$ , compute their dot product and norms. Are they orthogonal?
- 4 Find the angle between  $(1, 0, 1)^T$  and  $(1, 1, 0)^T$ .

# Questions?